

ELENA PAGNIN

Assistant Professor at Chalmers
University of Technology

Personal Info

Nationality: Italian, Swedish

E-mail: elenap@chalmers.se

Webpage: epagnin.github.io

Profiles: Google Scholar, LinkedIn
Scopus

ORCID: 0000-0002-7804-6696

Impact (Google Scholar)

No. publications: **30+**

No. citations: **500+**

h-index: **13**

Education & Titles

since 2022	Assistant Professor Tenure-Track (Forskarassistent)	Chalmers (SE)
2020 - 2022	Associate Senior Lecturer (Biträdande Apr-Aug Universitetslektor)	Lund University (SE)
2019 - 2020	Post-Doctoral Researcher in Cryptography Feb - Mar	Aarhus University (DK)
2014 - 2019	Ph.D. in Computer Science (Cryptography) May - Jan Thesis Title: " <i>Be more and be merry: enhancing data and user authentication in collaborative settings</i> "  Supervisor: A. Sabelfeld	Chalmers (SE)
2016	Licentiate Degree of Engineering Aug 25th Thesis Title: " <i>Authentication under Constraints</i> "  Supervisor: A. Mitrokotsa	Chalmers (SE)
2012 - 2013	Project Officer (paid Researcher Assistant position) Aug - Feb Master thesis project under Prof. F. Oggier supervision	Nanyang TU (SG)

Recent Achievements (Grants & Awards)

2025	Nominated for the Pedagogical Prize 2025 by the IT students; and my Cryptography course (TDA352/DIT352) appears in the recommended list of educational resources of campus totalforsvar	
2023-2024	Parental Leave (10 months)	
2023	Co-PI in a seed funding awarded by Chalmers' Areas of Advance on the topic <i>Towards a Multi-Layer Security Vision for Transportation Systems in the 6G Era</i>	(0.5M SEK)
2023-2027	PI of the prestigious Starting Grant awarded by the Swedish Research Council (VR starting) on <i>Progressive Verification of Cryptographic Schemes</i>	(4M SEK)
2020-2022	PI in a personal grant awarded to strategic research areas by the Excellence Center ELLIIT	(1.6M SEK)
2020	Best Paper Award for the full-version of " <i>Multi-Key Homomorphic Authenticators</i> " (extended abstract appeared in ASIACRYPT16) granted by the IET Information Security Journal. [This kind of Premium Award is given to recognize the best research papers published during the last two years]	

Supervision of PhD Students (= I am the main advisor, = I co-supervise together with)

2024 - now	Lucia Lavagnino : <i>Transparency Logs</i> () Planned PhD Defense: Nov 2029	Chalmers (SE)
2023 - now	Adrian Perez Keilty : <i>Progressive Verification of Cryptographic Schemes</i> () Planned PhD Defense: 20 Aug 2028	Chalmers (SE)
2023 - now	Hanna Ek : <i>Advanced Properties for Digital Signatures</i> () Planned PhD Defense: 31 Dec 2027	Chalmers (SE)
2019 - now	Ivan Oleynikov : <i>Privacy-Preserving Location Proximity Testing</i> () Licentiate: 30 Sept 2022. Planned PhD Defense: December 2025	Chalmers (SE)
2022 - now	Arthur Nijdam : <i>Secure Machine Learning for the Medical Sector</i> ( P. Stankovski-Wagner) Planned PhD Defense: September 2027	Lund University (SE)

Graduated Students

2025.03.07	Joakim Brorsson : <i>On the Trustworthiness of Trusted Third Parties</i> ( P. Stankovski-Wagner)	Lund University (SE)
2023.03.21	Martin Gunnarsson : <i>Securing IoT Systems</i> ( Prof. C. Gehrman)	Lund University (SE)
2023.02.28	Hadi Sehat : <i>Dual Deduplication in multi-client setting and its applications</i> ( Prof. D. Lucani)	Aarhus University (DK)

Selection of Professional Activities

09.06.2025	Member of the grading committee for the PhD Defence of Joel Gärtner on "Lattice-Based Post-Quantum Cryptography and Quantum Algorithms"	KTH (SE)
31.01.2024	Contributed to the referral to a new National law proposal on a Swedish electronic identification system <i>En säker och tillgänglig statlig e-legitimation</i> (SOU 2023:61)	
since 2021	Program Committee Member for: EuroCryp26, LatinCryp25, ACNS25, LatinCryp23, ACNS23, SEC@SAC22, ACISP21.	
since 2022	Mentor in the WISE-WWACQT mentorship program	
11.10.2023	Seminar speaker at ICT Area of Advance full-day seminar on Navigating the Cybersecurity Landscape on <i>Fortifying the Digital Fortress: Provably Secure Cryptography</i>	Chalmers (SE)

2023	Invited Speaker at AI NORDIC POWWOW on “Cybersecurity and AI for Company Security”	Lund (SE)
2023	Seminar speaker at the BARC center on <i>Progressive Verification for Cryptographic Schemes</i>	KU Copenhagen (DK)
2021-2022	Scientific leader at the EIT Department for the SSF “SMARTY” grant RIT17-0035 (22M SEK)	Lund University (SE)
2022	Invited Speaker at the seminar series at <i>Protocol Labs: Extended Threshold Ring Signatures</i>	(remote)
2022	Invited Speaker at the CRC seminar series at <i>TII Research Centers: Progressive and Efficient Verification</i>	(remote)
2022	Invited Speaker at a National hearing of researchers on “Innovative Processes for Data Integrity and Data Sharing” organized by the Swedish Ministry for Integrity Protection (Integritetsskyddsmyndigheten - IMY)	Stockholm (SE)
2022	Invited Panelist on “Allyship and Inclusion” at CRYPTO22	Santa Barbara (CA, USA)
2021	Invited Speaker at ELLIIT initiative on <i>Future-Oriented Research: “Enhancing Data Authentication”</i>	Lund (SE)
2021	Participant in the <i>Researchers’ Grand Prix: Security and Privacy in the Digital Era</i>	Helsinborg (SE)
2020	Invited Speaker at <i>Framtidsveckan: AI, digitalisering och integritet – vad får vi för vår hälsodata?</i> Presentation and Panel Discussion on “Contact tracing apps and their security dilemmas”	Lund (SE)

Teaching Experience

2015-now	Master’s Theses: Supervisor of 10, Examiner for 3.	Chalmers, Università di Milano, Lund University
2022-now	Course Responsible for the Masters level course <i>Cryptography</i> (TDA352/DIT352, 7.5hp)	Chalmers (SE)
2022	Lecturer and Co-instructor for the Masters level course <i>Advanced Cryptography</i> (EITN85, 7.5hp)	Lund University (SE)
2020 - 2022	Lecturer and Course Responsible for the Masters course <i>Advanced Web Security</i> (EITN41, 7.5hp)	Lund University (SE)
2021	Course Responsible for the PhD level course <i>Frontiers in Security Research</i> (EIT190F, 7.5hp)	Lund University (SE)

Selected List of Publications

CRYPTO25 	Keilty, Aranha, Elena Pagnin , and Rodríguez-Henríquez: “ <i>That’s AmorE: Amortized Efficiency for Pairing Delegation</i> ”. In: <i>Advances in Cryptology – CRYPTO</i> (2025).
PKC25 	Baum, David, Elena Pagnin , and Takahashi: “ <i>Universally Composable Interactive and Ordered Multi-Signatures</i> ”. In: <i>International Conference on Practice and Theory of Public-Key Cryptography</i> (2025).
AC24 	David, Engelmann, Frederiksen, Kohlweiss, Elena Pagnin , and Volkhov: “ <i>Updatable Privacy-Preserving Blueprints</i> ”. In: <i>Annual International Conference on Theory and Application of Cryptology and Information Security - ASIACRYPT</i> (2024).
SCN24 	Baum, David, Elena Pagnin , and Takahashi: “ <i>CaSCaDE:(Time-Based) Cryptography from Space Communications Delay</i> ”. In: <i>International Conference on Security and Cryptography for Networks</i> (2024).
EuroS&P24 	Nelson, Elena Pagnin , and Askarov: “ <i>Metadata Privacy Beyond Tunneling for Instant Messaging</i> ”. In: <i>IEEE European Symposium on Security and Privacy - EuroS&P</i> (2024).
CT-RSA23 	Brorsson, David, Gentile, Elena Pagnin , and Stankovski-Wagner: “ <i>PAPR: Publicly Auditable Privacy Revocation for Anonymous Credentials</i> ”. In: <i>The Cryptographers’ Track at RSA Conference</i> (2023).
ACNS22 	Boschini, Fiore, and Elena Pagnin : “ <i>Progressive and Efficient Verification for Digital Signatures</i> ”. In: <i>International Conference on Applied Cryptography and Network Security - ACNS</i> (2022).
PKC22 	Aranha, Hall-Andersen, Nitulescu, Elena Pagnin , and Yakoubov: “ <i>Count Me In! Extendability for Threshold Ring Signatures</i> ”. In: <i>International Conference on Practice and Theory of Public-Key Cryptography - PKC</i> (2022).
ESORICS20 	Oleynikov, Elena Pagnin , and Sabelfeld: “ <i>Where Are You Bob? Privacy-Preserving Proximity Testing with a Napping Party</i> ”. In: <i>European Symposium on Research in Computer Security – ESORICS</i> (2020).
AC16 	Fiore, Mitrokotsa, Nizzardo, and Elena Pagnin : “ <i>Multi-key Homomorphic Authenticators</i> ”. In: <i>Annual International Conference on Theory and Application of Cryptology and Information Security - ASIACRYPT</i> (2016).